DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Subject :Computer Organization and Microprocessors

Class: B.E (CSE)
Semester: III
Academic Year: 2025-2026

Max Marks: 30
Duration : 1 1/2
Date: 24-11-2025

		Marks	BT	CO
	PART A **Answer All Questions (5 x 2 = 10 M)**			
1. a)	Differentiate between Microprocessor and Microcontroller.	2	2	4
b)	Draw the structure of memory hierarchy.	2	2	5
c)	Brief about the internal data memory and program memory of 8051 Microcontroller.	2	2	4
d)	List out the pipeline conflicts with brief explanation.	2	2	3
e)	Compare between Memory mapped I/O and I/O mapped I/O.	2	2	5
	PART B **Answer Any Two Questions (2 x 10 = 20 M)**			
2.	a) Explain the operation of four-segment Instruction pipeline with suitable diagrams.	6	2	3
	b) Explain about Virtual memory.	4	2	2
3.	a) Draw the architecture of 8085 Microprocessor and explain the same in detail.	6	2	4
	b) Explain the addressing modes of 8051 Microcontroller.	4	2	4
4.	a) Write the program to display "WELCOME" in LCD by interfacing with 8051MC and draw the suitable hardware connections.	6	3	5
	b) What is DMA? Draw the block diagram of DMA Controller (8257).	4	2	5